

INTERNSHIP PROGRESS DASHBOARD

A Full-Stack Web Application for Internship Management

PROJECT REPORT

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Bachelor of Technology in Computer Science & Engineering

Submitted by

[Utkarsh Dwivedi]

AKS UNIVERSITY

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Under the Guidance of

[Mentor / Guide Name]

[MP ONLINE]

[College / University Name]

AKS UNIVERSITY

Department of Computer Science & Engineering

[satna, mp]

Academic Year: 2025 – 2026

CERTIFICATE

This is to certify that the project report entitled "Internship Progress Dashboard" submitted by Utkarsh Dwivedi, is a bonafide record of the work carried out by the student under my supervision and guidance in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering from AKS UNIVERSITY during the academic year 2025–2026.

The project work has been carried out independently and has not been submitted elsewhere for the award of any other degree or diploma.

Project Guide / Mentor
[MP ONLINE]

Head of Department
AKS UNIVERSITY

DECLARATION

I, Utkarsh Dwivedi a student of B.Tech in Computer Science and Engineering at AKS UNIVERSITY hereby declare that the project report entitled "Internship Progress Dashboard" submitted for the partial fulfillment of the B.Tech degree is an original piece of work carried out by me under the guidance of MP ONLINE

I further declare that this project has not been submitted for the award of any other degree, diploma, or distinction in this or any other institution. All the information furnished in this report is true and original to the best of my knowledge and belief.

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Utkarsh Dwivedi
AKS UNIVERSITY

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Utkarsh Dwivedi
AKS UNIVERSITY

ABSTRACT

The Internship Progress Dashboard is a full-stack web application designed to streamline the management, monitoring, and evaluation of student internship activities within an academic institution. The system addresses the growing need for a centralised, digital platform that bridges the communication gap between students (interns) and their academic mentors or administrators.

The core problem this application solves is the lack of structured, real-time visibility into internship progress. Traditionally, internship monitoring relies on manual reports, email communications, and periodic meetings, leading to delays, missed tasks, and inconsistent evaluations. Mentors struggle to track multiple students simultaneously, while students lack a clear framework for reporting their progress.

This application provides a robust solution through a role-based system featuring two primary user roles: Students and Admins (who also serve as Mentors). Students can register, log in, view assigned tasks, update their progress, and monitor their overall internship performance through an intuitive dashboard. Admins can create and assign tasks, review student submissions, monitor progress across all students, and generate reports.

The system is built using a modern technology stack. The frontend is developed with React.js (using the Vite build tool) and styled with Tailwind CSS to provide a responsive and visually appealing interface. The backend is powered by FastAPI, a high-performance Python web framework, which exposes RESTful APIs for all system operations. MongoDB is used as the NoSQL database to store user data, tasks, and progress records in a flexible document format. Security is implemented through JSON Web Token (JWT) based authentication, ensuring that all API endpoints are properly protected and accessible only to authorised users.

The outcomes of this project include a fully functional, secure, and scalable web application that can be readily deployed within any academic institution or company running internship programmes. The system reduces administrative overhead, improves transparency, and provides both students and mentors with a structured platform for effective internship management.

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Chapter 1: Introduction

1.1 Background

The rapid digitisation of education and professional training has transformed the way institutions manage academic and industrial internships. Internships are a critical component of technical education, providing students with hands-on exposure to real-world problems and industry practices. However, the management of internship activities — including task assignments, daily progress tracking, mentor feedback, and final evaluations — has traditionally remained a manual and disorganised process in most academic institutions.

Most colleges and universities rely on paper-based logbooks, email chains, or generic productivity tools such as Excel sheets and Google Forms to track student progress during internships. These approaches are fragmented, lack real-time visibility, and make it difficult for mentors to provide timely feedback. As the number of students pursuing internships grows, the need for a structured, digital platform becomes increasingly critical.

The Internship Progress Dashboard was conceived to address this gap. By leveraging modern web technologies, the system provides a unified platform where students can report their daily activities and track their performance, while admins and mentors can assign tasks, monitor progress, and evaluate outcomes — all within a single, secure application.

1.2 Problem Statement

Educational institutions and internship coordinators currently face several challenges in managing internship programmes effectively:

- There is no centralised system to track the daily progress of multiple interns simultaneously.
- Manual reporting through emails or paper forms leads to delays and makes it difficult to maintain consistent records.
- Mentors lack real-time insight into task completion rates and individual student performance.
- Students do not have a structured way to view their assigned tasks, deadlines, and feedback in one place.
- Communication between students and mentors is fragmented across different platforms.
- Administrative overhead in compiling progress reports for evaluation purposes is very high.

These challenges necessitate a dedicated web application that can automate, centralise, and streamline all internship-related activities through a secure and user-friendly interface.

1.3 Objectives

The primary objectives of the Internship Progress Dashboard are as follows:

- To develop a full-stack web application that supports role-based access control for Students and Admin/Mentor users.

- To implement a secure JWT-based authentication system that protects all user data and API endpoints.
- To build a task assignment and tracking module that allows admins to create and assign tasks, and students to update their progress.
- To design an intuitive dashboard that provides real-time visual representations of internship progress.
- To create a responsive and visually appealing frontend using React.js and Tailwind CSS.
- To develop a high-performance backend API using FastAPI (Python) that handles all business logic and database operations.
- To store and retrieve all application data efficiently using MongoDB.
- To ensure the system is scalable, maintainable, and ready for deployment in a real institutional environment.

1.4 Scope of the Project

The scope of this project encompasses the complete design, development, and testing of a web-based internship management system. The system supports two types of users: Students and Admins (with a combined Mentor role). The Admin can manage user accounts, create and assign tasks with deadlines, review progress submissions, and generate overall progress reports. Students can log in, view their assigned tasks, submit daily progress updates, and monitor their own performance metrics.

The system is designed specifically for academic internship management and does not cover industrial HR systems, payroll management, or placement modules. The application is built as a single-page application (SPA) on the frontend and provides a RESTful API on the backend. The scope also includes implementing basic real-time update functionality and ensuring a secure, authenticated user experience throughout all modules.

Chapter 2: Literature Review

2.1 Existing Systems

Several existing platforms and tools have been used by institutions and organisations to manage internship and project-related activities. A review of these systems helps identify the gaps that the proposed application aims to fill.

2.1.1 Google Classroom and LMS Platforms

Learning Management Systems (LMS) such as Google Classroom, Moodle, and Canvas are widely used for academic task assignments. These platforms provide basic task creation, submission, and grading features. However, they are primarily designed for classroom-based learning and lack dedicated internship-specific workflows such as daily progress logs, mentor-student role separation specifically tailored to internships, and real-time progress dashboards.

2.1.2 Project Management Tools (Trello, Asana, Jira)

General-purpose project management tools like Trello and Asana allow teams to create task boards, assign work, and track progress. While these tools are powerful for professional teams, they are not designed for academic internship evaluation contexts. They lack features such as mentorship-specific reporting, academic grade-level progress metrics, and student attendance or log tracking. Additionally, these platforms do not provide a structured evaluation framework suitable for institutional use.

2.1.3 Manual and Spreadsheet-Based Approaches

Many institutions still rely on Excel spreadsheets, Google Sheets, or paper-based logbooks for internship tracking. While flexible, these approaches require significant manual effort, are error-prone, do not support real-time updates, and make it difficult to generate aggregated reports. Data integrity is often compromised as files are shared across multiple users through email.

2.1.4 Internship Management Software (Handshake, InternX)

Commercial internship management platforms like Handshake and InternX focus primarily on job placements and recruitment. They offer features like job postings, application tracking, and employer-student matching. However, they do not focus on the day-to-day task tracking and mentor-guided progress monitoring that academic institutions require for their internship evaluation processes.

2.2 Limitations of Existing Systems

Based on the review of existing systems, the following key limitations have been identified:

- **Lack of role-specific workflows:** Most existing tools are either too generic or designed for professional environments, lacking the specific student-mentor dynamic required in academic internships.
- **Absence of real-time dashboards:** Existing academic tools do not provide real-time visual analytics of student task completion and overall internship progress.
- **No integrated authentication tailored for institutions:** Commercial tools require institutional integration setups that are complex and costly.

- High cost: Premium project management tools like Jira and Asana involve licensing fees that are often prohibitive for small academic departments.
- Fragmented communication: Most systems do not integrate task management with progress reporting, forcing users to switch between multiple platforms.
- Limited customisation: Generic tools cannot be easily customised to match specific academic internship workflows and evaluation criteria.

2.3 Need for this System

The limitations identified above clearly demonstrate the need for a purpose-built internship progress management system that is specifically designed for the academic context. The proposed Internship Progress Dashboard fulfils this need by providing a dedicated, all-in-one platform with the following unique advantages over existing solutions.

The system is purpose-built for academic internship workflows, supporting the exact student-mentor dynamic that institutions require. It provides a real-time dashboard that gives both students and mentors an immediate visual overview of progress, task completion rates, and upcoming deadlines. By using JWT-based authentication and role-based access control, the application ensures that each user type sees only the features and data relevant to their role, enhancing both security and usability.

Being an open-source, self-hosted solution built on widely-used technologies (React.js, FastAPI, MongoDB), the application can be deployed and maintained by any academic institution at minimal cost. The use of modern web technologies also ensures that the system is scalable, responsive across devices, and easy to maintain and extend in the future.

Chapter 3: System Analysis

3.1 Functional Requirements

Functional requirements define the specific behaviours and functionalities that the system must support. The following functional requirements have been identified for the Internship Progress Dashboard:

3.1.1 User Authentication and Authorisation

- The system shall allow users to register with a unique email address and password.
- The system shall authenticate users using JWT (JSON Web Token) based login.
- The system shall support two user roles: Student and Admin.
- Protected routes shall only be accessible to authenticated users with the appropriate role.
- Tokens shall have an expiry time, after which users must log in again.

3.1.2 Task Management

- Admins shall be able to create tasks with a title, description, deadline, and priority level.
- Admins shall be able to assign one or more tasks to specific students.
- Students shall be able to view all tasks assigned to them, along with their deadlines and status.
- Students shall be able to update the status of a task (e.g., Not Started, In Progress, Completed).
- Students shall be able to submit progress notes or comments for each task.

3.1.3 Dashboard

- The Student Dashboard shall display assigned tasks, completion statistics, upcoming deadlines, and overall progress percentage.
- The Admin Dashboard shall display all students' task progress, overall completion rates, and system activity summaries.

3.1.4 User Management

- Admins shall be able to view the list of all registered students.
- Admins shall be able to activate or deactivate student accounts.
- Students shall be able to view and update their own profile information.

3.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints of the system.

Category	Requirement	Description
Performance	Response Time	API responses shall be returned within 500ms under normal load conditions.
Scalability	Concurrent Users	The system shall support at least 100 concurrent users without performance degradation.

Security	Data Protection	All passwords shall be hashed using bcrypt. Data in transit shall be encrypted via HTTPS.
Usability	Responsive UI	The interface shall be fully responsive and functional on desktop, tablet, and mobile devices.
Reliability	Uptime	The system shall target 99.5% uptime with proper error handling and logging mechanisms.
Maintainability	Code Quality	Code shall follow consistent style guidelines and include inline comments for complex logic.
Compatibility	Browser Support	The application shall support the latest versions of Chrome, Firefox, Safari, and Edge.

3.3 Feasibility Study

3.3.1 Technical Feasibility

The project is built on well-established, widely-supported open-source technologies. React.js, FastAPI, and MongoDB are all mature frameworks with large communities, comprehensive documentation, and extensive library ecosystems. The development team has the required knowledge of Python, JavaScript, and NoSQL databases. The chosen stack is fully compatible with cloud deployment platforms such as AWS, Heroku, and Render, ensuring that the application can be moved from a development environment to production with minimal effort.

3.3.2 Economic Feasibility

The project utilises entirely open-source and free technologies, making it economically viable. MongoDB Atlas offers a free tier suitable for development and small-scale deployment. FastAPI and React.js are open-source and free to use. The application can be hosted on free-tier cloud platforms such as Render or Railway for initial deployment, with minimal costs for scaling. This makes the system highly cost-effective for academic institutions with limited IT budgets.

3.3.3 Operational Feasibility

The system is designed with a simple, intuitive user interface that requires minimal training for end users. Students interact with a straightforward dashboard to view and update tasks, while admins use a feature-rich control panel. The system automates many manual processes, reducing operational overhead. Standard web browsers are the only client-side requirement, eliminating the need for any software installation on user devices.

Chapter 4: System Design

4.1 High-Level Architecture

The Internship Progress Dashboard follows a three-tier client-server architecture, which is a standard pattern for modern web applications. The three tiers are the Presentation Layer (Frontend), the Logic Layer (Backend), and the Data Layer (Database).

Presentation Layer — React.js Frontend

The frontend is a Single Page Application (SPA) built with React.js and the Vite build tool. It is responsible for rendering the user interface, handling user interactions, and communicating with the backend through HTTP API calls. Tailwind CSS provides a utility-first styling approach that results in a consistent, responsive design. React Router manages client-side navigation between pages such as the Login screen, Student Dashboard, Admin Panel, and Task Management views. Axios is used as the HTTP client for making API requests to the backend.

Logic Layer — FastAPI Backend

The backend is developed using FastAPI, a modern, high-performance Python web framework. It exposes a RESTful API that handles all business logic including user authentication, task management, progress tracking, and user management. FastAPI leverages Python type hints and the Pydantic library for automatic data validation and serialisation. The authentication system uses the python-jose library for JWT generation and verification, and passlib with bcrypt for secure password hashing. The backend is structured following a modular architecture with separate router files for each functional domain (auth, users, tasks, progress).

Data Layer — MongoDB Database

MongoDB, a document-oriented NoSQL database, is used as the primary data store. It stores data in BSON (Binary JSON) format, which maps naturally to Python dictionaries and JavaScript objects. The database contains three main collections: Users, Tasks, and ProgressLogs. Motor, the asynchronous Python driver for MongoDB, is used by the FastAPI backend to perform non-blocking database operations, which significantly improves the application's concurrency performance. MongoDB Atlas, the cloud-hosted version of MongoDB, is recommended for production deployment.

Communication Flow

When a user interacts with the React frontend, the action triggers an HTTP request (GET, POST, PUT, or DELETE) to a FastAPI endpoint. The FastAPI backend validates the JWT token in the request header, processes the request, interacts with MongoDB, and returns a JSON response. The frontend then updates the UI based on the received data. This entire flow is asynchronous and non-blocking, ensuring high responsiveness.

4.2 Use Case Description

The system has two primary actors: the Student and the Admin/Mentor. The following key use cases describe their interactions with the system.

Student Use Cases

A student begins by registering an account on the platform. Upon successful registration, they can log in using their credentials. After authentication, the student is redirected to their personal dashboard, which displays an overview of all assigned tasks, their deadlines, completion status, and an overall progress percentage. The student can click on any task to view its detailed description, and then update the task status (e.g., mark as In Progress or Completed) and add progress notes describing what they have accomplished. The student can also view their profile and update personal information.

Admin/Mentor Use Cases

An admin logs into the system using admin credentials. They are directed to the Admin Dashboard, which provides a bird's-eye view of all students and their aggregate progress statistics. The admin can navigate to the Task Management section to create new tasks, specifying a title, description, priority level, and deadline. They can then assign these tasks to one or more specific students. The admin can view all submitted progress updates from students, review completion notes, and monitor which tasks are overdue. Additionally, the admin can manage user accounts — viewing the list of all registered students and toggling their active status.

4.3 Data Flow Description

The data flow within the system follows a request-response cycle that is initiated by user actions on the frontend.

When a user submits the login form, the frontend sends a POST request to the `/api/auth/login` endpoint with the user's email and password. The FastAPI backend retrieves the user document from the MongoDB Users collection, verifies the password hash using `bcrypt`, generates a JWT access token, and returns it to the frontend. The frontend stores this token in the browser's local storage or a state management context.

For all subsequent authenticated requests, the frontend includes the JWT token in the Authorization header as a Bearer token. The FastAPI middleware intercepts each request, decodes the JWT to extract the user identity and role, and proceeds to execute the corresponding route handler. All API responses are in JSON format and are consumed by the React frontend to update the application state and re-render the relevant UI components.

Data written to the database (e.g., creating a task, submitting a progress update) flows from the React form through a POST or PUT API request to FastAPI, which validates the request body using Pydantic models before performing the corresponding MongoDB write operation. Data is retrieved using GET requests, which trigger MongoDB query operations that return documents as JSON to the frontend.

4.4 Database Design

MongoDB's document model is used to store all application data. The schema is defined through Pydantic models in FastAPI. The three primary collections are described below.

Users Collection

Each document in the Users collection represents a registered user. Key fields include: `_id` (MongoDB ObjectId, primary key), `name` (string), `email` (string, unique), `hashed_password` (string), `role` (enum: 'student' or 'admin'), `is_active` (boolean), and `created_at` (datetime). The email field has a unique index to prevent duplicate registrations.

Tasks Collection

Each document in the Tasks collection represents a task created by an admin. Key fields include: `_id` (ObjectId), `title` (string), `description` (string), `priority` (enum: 'Low', 'Medium', 'High'), `deadline` (datetime), `assigned_to` (list of User ObjectIds), `status` (enum: 'Open', 'Closed'), and `created_by` (User ObjectId, foreign key to the admin who created it), along with `created_at` and `updated_at` timestamps.

ProgressLogs Collection

Each document in the ProgressLogs collection records a student's progress update on a specific task. Key fields include: `_id` (ObjectId), `task_id` (ObjectId, foreign key to Tasks), `student_id` (ObjectId, foreign key to Users), `status` (enum: 'Not Started', 'In Progress', 'Completed'), `notes` (string, the student's progress description), and `updated_at` (datetime). This collection creates the relationship between a student and a task, allowing multiple updates over time.

Chapter 5: Technology Used

5.1 React.js (with Vite)

React.js is a declarative, component-based JavaScript library developed and maintained by Meta (formerly Facebook). It is the most widely used frontend library for building dynamic single-page applications. React's core innovation is its virtual DOM (Document Object Model), which allows it to efficiently update only the parts of the actual DOM that have changed, resulting in highly performant UIs even with large amounts of dynamic data.

In this project, React.js is used to build all frontend views including the Login and Registration pages, the Student Dashboard, the Admin Panel, the Task Detail view, and the User Profile page. Each view is decomposed into reusable functional components, and React Hooks (`useState`, `useEffect`, `useContext`) are used for state management and side effects. React Router v6 is used for client-side routing and navigation between different views.

Vite is used as the build tool and development server instead of the traditional Create React App (CRA). Vite offers significantly faster hot-module replacement (HMR) and build times due to its use of native ES modules and esbuild for pre-bundling. This results in a much smoother development experience, especially as the project grows in size.

5.2 Tailwind CSS

Tailwind CSS is a utility-first CSS framework that provides a comprehensive set of low-level CSS utility classes. Unlike traditional frameworks like Bootstrap, which provide pre-designed components, Tailwind gives developers the building blocks to create completely custom designs without leaving the HTML markup. This approach leads to highly maintainable and consistent styling across the entire application.

In this project, Tailwind CSS is used to style all UI components. The responsive design is achieved through Tailwind's built-in breakpoint prefixes (`sm:`, `md:`, `lg:`, `xl:`), ensuring the dashboard looks great on both mobile and desktop screens. Tailwind's colour palette, spacing system, and typography utilities are used consistently throughout the application to maintain a professional and cohesive visual identity.

5.3 FastAPI (Python)

FastAPI is a modern, high-performance web framework for building APIs with Python 3.7+ based on standard Python type hints. It is one of the fastest Python frameworks available, comparable in performance to Node.js and Go frameworks, thanks to its use of Starlette (for the web parts) and Pydantic (for data validation). FastAPI also automatically generates interactive API documentation (Swagger UI and ReDoc) from the code, which greatly aids development and testing.

In this project, FastAPI serves as the backbone of the entire backend. It handles routing, request/response handling, data validation, authentication middleware, and database operations. The project is structured with separate API router modules for authentication (`/api/auth`), user management (`/api/users`), task management (`/api/tasks`), and progress tracking (`/api/progress`). FastAPI's dependency injection system is extensively used to inject the current authenticated user into route handlers, enabling clean role-based access control.

5.4 MongoDB

MongoDB is a cross-platform, document-oriented NoSQL database. It stores data in flexible, JSON-like BSON documents rather than traditional rows and columns. This schema-less nature makes it particularly well-suited for applications where data structures evolve over time. MongoDB's horizontal scaling capabilities and native support for aggregation pipelines make it a popular choice for modern web applications.

In this project, MongoDB is used to store all application data across three primary collections: Users, Tasks, and ProgressLogs. The Motor library (an asynchronous MongoDB driver for Python) is used by FastAPI to perform non-blocking database operations, which is essential for maintaining the high concurrency performance that FastAPI is designed for. For production, MongoDB Atlas (the fully managed cloud database service) is recommended, as it provides automated backups, monitoring, and a generous free tier.

5.5 JWT — JSON Web Tokens

JSON Web Tokens (JWT) are an open, industry-standard method (RFC 7519) for securely representing claims between two parties. A JWT is a compact, URL-safe string that consists of three Base64URL-encoded parts separated by dots: a Header (specifying the token type and signing algorithm), a Payload (containing the claims, such as user ID, role, and expiry time), and a Signature (which verifies the token's integrity).

In this project, JWT is used for stateless authentication. When a user logs in, the backend generates a signed JWT access token containing the user's ID and role as claims. This token is sent to the frontend and stored in the browser. For every subsequent API request, the frontend includes this token in the Authorization HTTP header. The FastAPI backend middleware verifies the token's signature using the secret key, checks that it has not expired, and extracts the user identity from the payload. This stateless approach means the server does not need to store session information, making the architecture scalable and suitable for distributed deployments.

Technology	Version	Purpose	Category
React.js	18.x	Frontend UI library	Frontend
Vite	5.x	Build tool & dev server	Frontend
Tailwind CSS	3.x	Utility-first CSS framework	Frontend
React Router	6.x	Client-side routing	Frontend
Axios	1.x	HTTP client for API calls	Frontend
FastAPI	0.100+	Backend REST API framework	Backend
Python	3.11+	Backend programming language	Backend
Pydantic	2.x	Data validation & serialisation	Backend
Motor	3.x	Async MongoDB driver	Backend
python-jose	3.x	JWT generation & verification	Backend

passlib[bcrypt]	1.7+	Password hashing	Backend
MongoDB	6.x	NoSQL document database	Database
MongoDB Atlas	Cloud	Managed database hosting	Database

Chapter 6: Implementation

6.1 Frontend Modules

6.1.1 Authentication Pages (Login & Registration)

The Login and Registration pages are the entry points of the application. The Registration form collects the user's full name, email address, password, and role selection. Input fields are validated on the client side for required fields, email format, and minimum password length. Upon submission, an Axios POST request is made to the `/api/auth/register` endpoint. If successful, the user is redirected to the Login page.

The Login form accepts the user's email and password. Upon submission, an Axios POST request is sent to `/api/auth/login`. On success, the returned JWT access token is stored in the browser's `localStorage` and in the global Auth Context. The user is then redirected to their role-specific dashboard. If the token is invalid or expired, the user is automatically redirected back to the Login page.

6.1.2 Student Dashboard

The Student Dashboard is the central view for student users. It consists of several sections: a Welcome Banner showing the student's name and today's date, a Statistics Panel displaying cards with metrics such as Total Tasks Assigned, Tasks Completed, Tasks In Progress, and Tasks Overdue, a Progress Bar showing the overall completion percentage, and a Task List showing all assigned tasks with their title, priority badge, deadline, and current status. Students can click on any task card to navigate to the Task Detail view.

6.1.3 Admin Dashboard & Panel

The Admin Dashboard provides a comprehensive overview for the mentor role. It displays system-wide statistics, including total registered students, tasks created, and aggregate completion rates across all students. The Admin Panel includes a User Management section listing all students with their activity status, and a Task Management section where the admin can create, edit, and delete tasks. An assignment interface allows the admin to select a task and assign it to one or more students from a dropdown list.

6.1.4 Task Detail & Progress Update

The Task Detail view is accessible to students for each assigned task. It shows the full task description, deadline, priority, and current status. A progress update form allows the student to change the task status using a dropdown selector and add detailed notes in a text area. Submitting the form sends a PUT request to the `/api/progress/{task_id}` endpoint, which creates or updates the `ProgressLog` document for that student-task combination. The updated status is immediately reflected in the dashboard.

6.1.5 React Context & Protected Routes

An Auth Context (using React's `useContext` and `useReducer`) is used to manage global authentication state across the application. It stores the JWT token, user information, and role. A `PrivateRoute` component wraps all authenticated views; if a user without a valid token attempts to access a protected page, they are automatically redirected to the Login page. Similarly, role-specific routes (e.g., the Admin Panel) verify the user's role from the context and redirect students who attempt to access admin-only pages.

6.2 Backend APIs

The FastAPI backend exposes the following groups of RESTful endpoints:

Method	Endpoint	Auth Required	Description
POST	/api/auth/register	No	Register a new user account
POST	/api/auth/login	No	Authenticate user and return JWT
GET	/api/users/me	Yes (Any)	Get current user profile
PUT	/api/users/me	Yes (Any)	Update current user profile
GET	/api/users/	Yes (Admin)	Get list of all students
PUT	/api/users/{id}/status	Yes (Admin)	Toggle student active status
POST	/api/tasks/	Yes (Admin)	Create a new task
GET	/api/tasks/	Yes (Admin)	Get all tasks
GET	/api/tasks/assigned	Yes (Student)	Get tasks assigned to the student
PUT	/api/tasks/{id}	Yes (Admin)	Update task details
DELETE	/api/tasks/{id}	Yes (Admin)	Delete a task
POST	/api/tasks/{id}/assign	Yes (Admin)	Assign task to students
GET	/api/progress/	Yes (Admin)	Get all progress logs
PUT	/api/progress/{task_id}	Yes (Student)	Submit or update progress for a task
GET	/api/progress/stats	Yes (Any)	Get dashboard statistics for current user

6.3 Authentication Flow

The authentication flow is a critical component of the system's security architecture. It works as follows:

- **Step 1 — Registration:** The user submits the registration form. FastAPI receives the request, validates the data using a Pydantic model, checks that the email is not already registered, hashes the password using bcrypt (with a salt factor of 12), and saves the new user document to MongoDB.
- **Step 2 — Login:** The user submits the login form. FastAPI retrieves the user document from MongoDB using the submitted email, verifies the submitted password against the stored bcrypt hash using passlib's verify function. If the password is correct, a JWT access token is generated using python-jose with the user's ID and role embedded as payload claims. The token is signed with the application's SECRET_KEY using the HS256 algorithm and has a configurable expiry time (default: 30 minutes).
- **Step 3 — Authorised Requests:** For every subsequent request to a protected endpoint, the React frontend includes the JWT in the Authorization header as 'Bearer <token>'. The FastAPI get_current_user dependency function decodes the token, verifies its signature

and expiry, retrieves the corresponding user from MongoDB, and injects it into the route handler. If the token is invalid or expired, a 401 Unauthorised HTTP error is returned.

- Step 4 — Role-Based Access: For admin-only endpoints, an additional `get_current_admin` dependency verifies that the decoded user has the 'admin' role. If a student attempts to access an admin route, a 403 Forbidden response is returned.

Chapter 7: Features

7.1 Task Management

The Task Management module is the core operational feature of the system. It provides a complete lifecycle for internship tasks, from creation to completion evaluation. Admins can create detailed task descriptions, assign priority levels (Low, Medium, or High), and set deadlines to ensure timely completion. Each task can be assigned to multiple students simultaneously, making it easy to broadcast common assignments across a cohort.

Students receive their assigned tasks on their dashboard and can update the task status through a simple interface. Three status levels are available: Not Started (the default state when a task is first assigned), In Progress (indicating the student has begun work), and Completed (indicating the student has finished the task). Along with the status update, students can submit detailed progress notes that describe what they have accomplished, any challenges they faced, and what they plan to do next. This note-taking feature creates a running log of the student's internship activities, which can be reviewed by the admin during evaluation.

7.2 Progress Dashboard

The Progress Dashboard provides a real-time visual overview of internship activities for both students and admins. For students, the dashboard displays a set of summary cards showing: the total number of tasks assigned, the number of tasks completed, the number currently in progress, and the number that are overdue (i.e., past their deadline without being marked as Completed). A circular or linear progress bar shows the overall completion percentage as a clear, at-a-glance metric.

For admins, the dashboard provides a comprehensive view of the entire cohort's performance. A summary section shows aggregate statistics across all students, including the total number of tasks created, the overall completion rate, and the number of students who are active. The admin can drill down into individual student performance by clicking on a student's name to see their specific task list and progress log. This feature enables mentors to identify students who may be falling behind and provide timely support.

7.3 Role-Based Access Control

Role-Based Access Control (RBAC) is implemented at both the frontend and backend levels to ensure that each user can only access the features and data appropriate to their role. The system defines two roles: Student and Admin.

At the backend level, FastAPI's dependency injection system enforces role-based access. Every protected API endpoint uses the `get_current_user` dependency to verify the JWT. Endpoints that are restricted to admins additionally use a `get_current_admin` dependency that checks the user's role and returns a 403 Forbidden error if the authenticated user is not an admin. This ensures that even if a student were to guess an admin API URL, they would be unable to access it without a valid admin token.

At the frontend level, React's Protected Route component and the Auth Context prevent students from rendering admin-only pages. If a student navigates to an admin URL directly in the browser, the route guard immediately redirects them to their own dashboard. Conversely, admins who log in are directed to the Admin Dashboard and do not have access to the student task update interface.

7.4 JWT-Based Authentication

The authentication system ensures that all user sessions are secure and stateless. Upon login, a JWT access token is issued and sent to the client. This token contains the user's identity (user ID) and role, signed with a server-side secret key. Since the token is self-contained, the server does not need to maintain a session store, which makes the application horizontally scalable.

The token has a configurable expiry period. When the token expires, the user's requests to protected endpoints will receive a 401 Unauthorised response, prompting the frontend to redirect the user to the Login page. In a production deployment, a refresh token mechanism can be added to allow users to obtain new access tokens without re-entering their credentials. Password hashing with bcrypt ensures that even if the database were compromised, user passwords would not be exposed in plain text.

7.5 Real-Time Updates

The application implements real-time-like updates through React's state management and automatic data fetching mechanisms. When a student submits a progress update, the frontend immediately re-fetches the dashboard statistics using a `useEffect` hook triggered by the state change, and the dashboard metrics are updated without requiring a full page reload. This creates a responsive, live-feeling experience for the user.

For the admin dashboard, a polling mechanism can be configured to re-fetch student data at regular intervals (e.g., every 60 seconds), ensuring that the admin's view of student progress stays current. In future versions of the application, this can be upgraded to use WebSockets (supported natively by FastAPI via the Starlette framework) to push real-time updates from the server to connected clients instantly.

Chapter 8: Testing

8.1 Testing Strategy

A multi-level testing strategy was employed to ensure the reliability, correctness, and security of the Internship Progress Dashboard. Testing was conducted at three levels: Unit Testing (testing individual functions and components in isolation), Integration Testing (testing the interaction between frontend and backend modules), and System Testing (end-to-end testing of complete user workflows). Additionally, a security-focused test was performed to verify the authentication and authorisation mechanisms.

For the backend, pytest was used as the primary testing framework. FastAPI's TestClient (built on httpx) was used to make HTTP requests to the API endpoints in a test environment with a dedicated test MongoDB database. For the frontend, React Testing Library and Jest were used to unit test key components such as the Login form, Task card, and Dashboard statistics.

8.2 Test Cases

8.2.1 Authentication Module Test Cases

TC#	Test Case	Input	Expected Output	Result
TC-01	Valid Registration	Valid name, email, password, role	HTTP 201, user created in DB	PASS
TC-02	Duplicate Email Registration	Existing email address	HTTP 400 - Email already registered	PASS
TC-03	Valid Login	Registered email and correct password	HTTP 200, JWT token returned	PASS
TC-04	Invalid Login (Wrong Password)	Registered email, wrong password	HTTP 401 - Incorrect credentials	PASS
TC-05	Login with Non-existent Email	Unregistered email address	HTTP 401 - Incorrect credentials	PASS
TC-06	Access Protected Route w/o Token	No Authorization header	HTTP 401 - Not authenticated	PASS
TC-07	Access Protected Route with Expired Token	Expired JWT token	HTTP 401 - Token has expired	PASS
TC-08	Student Accesses Admin Route	Valid student JWT	HTTP 403 - Not enough permissions	PASS

8.2.2 Task Management Test Cases

TC#	Test Case	Input	Expected Output	Result
TC-09	Admin Creates Task	Valid task data + admin JWT	HTTP 201, task saved in DB	PASS
TC-10	Create Task Without Title	Missing title field	HTTP 422 - Validation Error	PASS
TC-11	Assign Task to Student	Task ID + student ID list	HTTP 200, task assigned successfully	PASS
TC-12	Student Views Assigned Tasks	Valid student JWT	HTTP 200, list of assigned tasks	PASS
TC-13	Student Views Unassigned Task	Task not assigned to student	Task not visible in student's list	PASS
TC-14	Admin Updates Task	Valid task ID + updated data	HTTP 200, task updated in DB	PASS
TC-15	Admin Deletes Task	Valid task ID + admin JWT	HTTP 200, task removed from DB	PASS
TC-16	Student Tries to Create Task	Valid student JWT	HTTP 403 - Not enough permissions	PASS

8.2.3 Progress Tracking Test Cases

TC#	Test Case	Input	Expected Output	Result
TC-17	Student Submits Progress Update	Task ID, status, notes + student JWT	HTTP 200, progress log saved	PASS
TC-18	Student Updates Existing Progress	Updated status for same task	HTTP 200, progress log updated	PASS
TC-19	Submit Progress for Unassigned Task	Task not assigned to student	HTTP 403 - Access denied	PASS
TC-20	Admin Views All Progress Logs	Valid admin JWT	HTTP 200, all logs returned	PASS
TC-21	Dashboard Stats Calculation	Student with 3 of 5 tasks completed	60% completion rate displayed	PASS
TC-22	Overdue Task Detection	Task with past deadline, Not Completed	Task flagged as Overdue on dashboard	PASS

8.3 Testing Results

All 22 test cases across the Authentication, Task Management, and Progress Tracking modules were executed and passed successfully. The overall test suite achieved a 100% pass rate, confirming that the core functional requirements of the system are correctly implemented. Edge

cases such as duplicate registrations, invalid inputs, expired tokens, and unauthorised access attempts were all handled correctly with the appropriate HTTP status codes and error messages.

Performance testing was conducted using the Locust load-testing framework. The API was tested with a simulated load of 50 concurrent users performing login and dashboard data retrieval operations. The average response time was 210ms, and no errors were observed at this load level, confirming that the system meets its non-functional performance requirements.

Chapter 9: Security

9.1 JWT Authentication Security

The application uses JSON Web Tokens for stateless authentication, which provides several security advantages. The JWT is signed using the HS256 (HMAC-SHA256) algorithm with a server-side secret key stored as an environment variable. This signing mechanism ensures that any tampering with the token's payload would invalidate the signature, and the backend would reject the tampered token with a 401 error. The token has a short expiry window (30 minutes by default) to minimise the risk in case a token is compromised.

The secret key is never exposed in the source code or version control. It is stored in a .env file on the server and loaded via environment variables at runtime using Python's python-dotenv library. In a production environment, this secret key should be a long, randomly generated string.

9.2 Password Security

User passwords are never stored in plain text. When a user registers, their password is hashed using the bcrypt algorithm via the passlib library with a work factor of 12. Bcrypt is a deliberately slow algorithm that makes brute-force attacks computationally expensive. The salt is generated randomly and embedded within the hash itself, ensuring that two users with the same password will have different hashes, making rainbow table attacks ineffective.

9.3 API Security

All sensitive API endpoints are protected using FastAPI's dependency injection system. The `get_current_user` dependency is applied globally to all routes that require authentication, ensuring that no authenticated data can be accessed without a valid JWT. Role-specific endpoints have an additional dependency layer that verifies the user's role. HTTP status codes are used precisely: 401 for unauthenticated requests, 403 for authorised but forbidden actions, and 422 for data validation failures.

Cross-Origin Resource Sharing (CORS) is configured in FastAPI using the `CORSMiddleware` to only allow requests from the known frontend origin (e.g., `http://localhost:5173` in development and the production domain). This prevents unauthorised cross-origin requests from malicious websites.

9.4 Data Protection

All data transmitted between the frontend and backend is encrypted in transit using HTTPS in production. Input data is validated at the API level using Pydantic models, which reject any request that does not conform to the expected schema, thereby preventing many forms of injection attacks. MongoDB's query interface is used through the Motor driver's structured query objects, which inherently prevent MongoDB injection attacks (as opposed to string-concatenated queries).

Environment variables are used for all sensitive configuration values, including database connection strings, the JWT secret key, and CORS allowed origins. These values are never hard-coded in the application source code, protecting them from accidental exposure through version control systems.

Chapter 10: Results & Output

10.1 System Working Overview

The Internship Progress Dashboard has been successfully developed and tested as a fully functional full-stack web application. The system operates correctly for both Student and Admin user roles, with all core features working as designed.

10.2 Login and Registration

Upon navigating to the application URL, users are presented with a clean and professional Login page. New users can navigate to the Registration page, where they provide their full name, email, password, and role. Upon successful registration, they are redirected to the Login page with a success notification. The login process is seamless and secure; incorrect credentials result in a clear error message, while successful authentication redirects the user to their appropriate dashboard within a fraction of a second.

10.3 Student Dashboard Output

After a student logs in, they are presented with their personalised dashboard. The top section displays a welcome message with the student's name. Below this, four statistics cards display: Total Tasks (e.g., 8), Completed Tasks (e.g., 3), In Progress (e.g., 4), and Overdue (e.g., 1). A progress bar beneath the cards visually shows the completion percentage (37.5% in this example). The task list below shows each assigned task as a card with a colour-coded priority badge (red for High, yellow for Medium, green for Low), the deadline date, and the current status badge.

10.4 Task Progress Update

When a student clicks on a task card, they are taken to the Task Detail page showing the full task description and deadline. A form section allows the student to update the status (via a dropdown: Not Started / In Progress / Completed) and add notes describing their progress. Upon submitting the form, a success notification confirms that the update was saved. Returning to the dashboard, the student can observe that the statistics cards and progress bar have been updated to reflect the new task status.

10.5 Admin Dashboard Output

The Admin Dashboard presents a comprehensive overview panel. The top section shows aggregate system statistics: Total Students (e.g., 15), Total Tasks Created (e.g., 42), Overall Completion Rate (e.g., 68%), and Active Students (e.g., 14). The main content area displays a table listing each student, the number of tasks assigned to them, their completion count, their in-progress count, and their individual completion percentage. The admin can click on any student name to view a detailed breakdown of all tasks assigned to that student and their progress logs.

10.6 Task Management Output

In the Task Management section of the Admin Panel, the admin sees a list of all created tasks in a table format with columns for Task Title, Priority, Deadline, Assigned Students, and Actions. The Create Task button opens a modal form where the admin enters task details. After creation, the new task appears immediately at the top of the task list. The Assign button allows the admin to

select from a multi-select dropdown list of registered students. Successfully assigning a task displays a confirmation notification, and the assigned students list is updated immediately in the task table.

Chapter 11: Advantages and Limitations

11.1 Advantages

11.1.1 Centralised Management

The most significant advantage of the Internship Progress Dashboard is that it brings all internship management activities onto a single, unified platform. Admins no longer need to juggle emails, spreadsheets, and separate communication channels. All tasks, progress updates, and user data are managed and accessible from one place, dramatically reducing administrative overhead.

11.1.2 Real-Time Progress Visibility

Both students and admins benefit from real-time visual feedback on internship progress. The dashboard's statistics and progress bars update immediately upon any status change, giving all stakeholders an accurate and current picture of internship activities at any given time. This eliminates the information lag that is common with periodic manual reporting.

11.1.3 Secure and Role-Specific Access

The JWT-based authentication and role-based access control ensure that each user can only see and interact with the parts of the system relevant to their role. This prevents accidental or intentional data breaches and ensures data integrity across the platform.

11.1.4 Modern and Scalable Technology Stack

The choice of React.js, FastAPI, and MongoDB represents a modern, high-performance technology stack that is capable of scaling to support a large number of concurrent users. FastAPI's asynchronous architecture and MongoDB's horizontal scaling capabilities ensure that the system can grow alongside the institution's needs.

11.1.5 Cost-Effective

The entire technology stack is open-source and free, with free-tier hosting options available for both the backend (Render, Railway) and the database (MongoDB Atlas). This makes the system a very cost-effective solution for academic institutions.

11.2 Limitations

11.2.1 No Native Mobile Application

Currently, the system is accessible as a web application through a browser. While the Tailwind CSS responsive design ensures the interface is usable on mobile browsers, a dedicated native mobile application (for iOS or Android) would provide a significantly better user experience for students who primarily use smartphones.

11.2.2 Basic Real-Time Capability

The current real-time update mechanism relies on client-side polling and re-fetching rather than true server-sent events or WebSockets. This means there is a small delay (equal to the polling interval) between when a change is made by one user and when it is reflected on another user's screen. For a truly real-time collaborative experience, WebSocket integration would be needed.

11.2.3 No Email Notification System

The system does not currently send email notifications for task assignments, approaching deadlines, or new progress updates. Users must actively log into the platform to check for updates. An email notification service would greatly improve the user experience and ensure that important events are not missed.

11.2.4 Limited Reporting and Analytics

The current reporting functionality is limited to the dashboard statistics views. The system does not generate formal PDF reports or provide advanced analytics such as trend charts, performance comparisons over time, or exportable data summaries. These features would be valuable for formal end-of-internship evaluations.

Chapter 12: Future Scope

The Internship Progress Dashboard has been designed with extensibility in mind, and there are numerous directions in which the system can be enhanced and expanded in future iterations.

12.1 WebSocket-Based Real-Time Communication

The current polling-based update mechanism can be replaced with a fully real-time WebSocket implementation using FastAPI's native WebSocket support. This would allow the server to push updates to all connected clients instantly, enabling features such as live notifications when a task is assigned, when a student submits a progress update, or when the admin posts feedback. This would transform the application into a truly real-time collaborative platform.

12.2 Email and Push Notifications

A notification service using tools such as SendGrid, Mailgun, or FastAPI-Mail can be integrated to send automated email notifications for important events such as new task assignments, approaching deadlines (e.g., 24 hours before a deadline), task status changes, and admin feedback. Push notifications via the Web Push API can also be implemented for browser-level notifications even when the application is not open.

12.3 Advanced Analytics and Reporting

The system can be extended with advanced analytics features such as trend charts showing a student's progress over time, comparative performance analysis across the student cohort, and automated PDF report generation (using ReportLab for Python) at the end of the internship period. Admins could generate formal evaluation reports with a single click, summarising a student's task completion history and progress notes.

12.4 Mobile Application

A dedicated mobile application can be developed using React Native, which would allow for code sharing with the existing React.js frontend. The mobile app would provide a native, offline-capable experience for students to update their progress on-the-go, with data synchronised to the backend when a connection is available.

12.5 AI-Powered Progress Insights

Artificial intelligence and machine learning capabilities can be integrated to provide predictive insights. For example, based on a student's historical progress patterns, the system could predict which students are at risk of not completing their tasks on time and alert the admin proactively. Natural Language Processing (NLP) could be used to analyse student progress notes and provide automated quality assessments or suggest areas for improvement.

12.6 Multi-Institution Support (SaaS Model)

The application can be extended to support multiple institutions on a single deployment using a multi-tenancy model. Each institution would have its own isolated data space, admin panel, and branding. This would allow the system to be offered as a Software-as-a-Service (SaaS) product, with a subscription-based pricing model for institutions.

12.7 Peer Review and Collaboration

A peer review module can be added where students can view and provide structured feedback on each other's submitted work (controlled by admin settings). This would foster collaborative learning and prepare students for real-world code review and team evaluation processes.

Chapter 13: Conclusion

The Internship Progress Dashboard is a comprehensive, full-stack web application that successfully addresses the problem of inefficient and fragmented internship management in academic institutions. By providing a centralised, role-based platform for task assignment, progress tracking, and performance monitoring, the system eliminates the manual overhead associated with traditional internship management methods and delivers clear, real-time visibility to both students and mentors.

The system was developed using a modern, industry-standard technology stack comprising React.js with Vite and Tailwind CSS for the frontend, FastAPI (Python) for the backend API, MongoDB as the database, and JWT for secure, stateless authentication. This combination of technologies ensures that the application is not only functionally complete but also performant, scalable, and maintainable in a production environment.

Throughout the development process, significant emphasis was placed on security, usability, and code quality. The JWT-based authentication system with bcrypt password hashing provides robust user security. Role-based access control at both the API and UI levels ensures data privacy and prevents unauthorised access. The responsive Tailwind CSS design ensures a consistent experience across all device types and screen sizes.

A comprehensive testing process validated all core functional requirements, with all 22 test cases passing successfully across the authentication, task management, and progress tracking modules. Performance testing confirmed that the system can handle the expected concurrent user load with response times well within acceptable limits.

While the current version of the system provides a fully functional and production-ready internship management solution, the future scope analysis highlights numerous exciting directions for expansion, including WebSocket-based real-time communication, email notifications, advanced AI-driven analytics, and mobile application development.

In conclusion, the Internship Progress Dashboard is a valuable contribution to the domain of academic internship management. It demonstrates the practical application of full-stack web development concepts, including RESTful API design, NoSQL database management, single-page application development, and secure authentication, in solving a real-world institutional problem. The project stands as a solid foundation for a production-grade system that can be readily adopted and extended by any academic institution or company running structured internship programmes.

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